

The Role of Algorithmic Audits and Other Soft Law Approaches in Informing Users' Calibrated Trust in Artificial Intelligence Tools

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Abstract

Artificial Intelligence (AI) is becoming more and more a part of our daily life and is affecting a wide variety of users. While there are many beneficial aspects of AI, there are also growing concerns about harms to the public. In addition, there is wide-spread distrust in AI in general in the public. Algorithmic audits could help to build calibrated trust by users. Algorithmic audits are slowly becoming more common, in particular in the Human Resource (HR) sector. Previous research has focused on the use and design of audits but less on how they are received by users' and how they could help users to calibrate their trust in AI-based systems to the trustworthiness of the system. The research described below aims to fill this gap. Based on initial work on algorithmic audits in the HR sector, the research will in the future expand to go beyond this sector and also cover other AI governance mechanisms, in particular soft law approaches such as the NIST AI Risk Management Framework.

CCS Concepts

• Social and professional topics → Professional topics → Management of computing and information systems → System management → Technology audits • Security and privacy → Formal methods and theory of security → Trust frameworks • Human-centered computing → Collaborative and social computing → Collaborative and social computing theory, concepts and paradigms → Computer supported cooperative work • Social and professional topics → Computing / technology policy → Government technology policy → Governmental regulations • Human-centered computing → Human computer

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interaction (HCI) → HCI design and evaluation methods → Field studies

Keywords

Doctoral Consortium, algorithmic auditing, trust in AI-based systems, AI auditors, AI users, qualitative field research

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1 INTRODUCTION

As Artificial Intelligence (AI) becomes increasingly common, it also raises more and more concerns in the public, and by users of AI. Multiple data bases collect data of AI gone wrong (e.g., <https://www.aiaaic.org/aiaaic-repository>). A recent global study with over 17,000 participants on Trust in AI found that over 70 % are concerned about potential risks of AI [1]. Regulation in the United States is sparse and reacts slowly in comparison to the speed AI develops. Multiple general non-binding guidelines exist, either as global frameworks or guidelines, or guidelines created by technology companies. However, concerns are raised on how such guidelines will be implemented and whether they are enough. One recent approach to increase calibrated trust in AI is through algorithmic audits. While the idea already came up about a decade ago [2] and audits have been common in other areas, like the financial sector, algorithmic audits are overall a relatively new concept. First regulation in the United States and Europe requires algorithmic audits, such as the New York City Law 144 for Automated Employment Decisions (AED) [3], or the upcoming EU AI Act [4]. These regulatory efforts have led to an increase in algorithmic audits and a growing interest in various forms of such audits. While most of the research concentrates on the kind of audits on the market and their effectiveness [5], less work has focused on users' perspectives on algorithmic audits.

2 RESEARCH QUESTIONS

My research explores how users trust in AI could be influenced by algorithmic audits and whether such audits could help build calibrated trust. It is based on the following research questions.

- RQ1: How much do users' of AI tools know about algorithmic audits?
- RQ2: How could algorithmic audits build calibrated trust in AI used in Human Resource?
- RQ3: How could algorithmic audits build calibrated trust in AI in general?
- RQ4: What other approaches, such as regulation or different soft law approaches (e.g. standards) could influence users' trust in AI?
- RQ5: How do hard law and soft law come together in building calibrated trust in AI?

3 METHODOLOGY

My previous studies have used a mixed-methods approach, analyzing interview and survey results. Initial work focused on an interview study with 19 auditing professionals in 2023 via Zoom, to gather insights about their perspectives on how to create trust in algorithmic audits. The interviews included auditors with varying degrees of experience as well as professionals creating auditing and certification standards. The interviews were followed by a large survey exploring users' trust in algorithmic audits in the Human Resource sector, surveying 436 users, 327 of them jobseekers, and 107 HR-managers and recruiters.

4 RESULTS

The foundation of my dissertation work is research in algorithmic audits in the Human Resource (HR) sector as this is an area that has been of particular interest of auditors. It also builds on previous research I conducted prior to my PhD studies exploring recruiters perspectives on the use of AI tools in HR [6]. Decisions in HR and especially regarding employment decisions are highly consequential and deserve special attention. The New York City Law is the first global regulation [7] requiring bias audits for automated employment decisions. The upcoming European AI Act [4] considers employment decisions as a high-risk algorithm and also requires assessments. These regulations show the importance of audits in the HR sector. My research builds on an interview study with AI-Auditing professionals exploring how they intend to create calibrated trust in algorithmic audits, and a survey distributed among jobseekers and HR-professionals on their perspectives regarding algorithmic audits in the HR sector and how such audits influence their trust in AI used in HR. The research was funded by a grant of the Notre Dame-IBM Tech Ethics lab which I received to study users' trust in Human Resource AI tools and AI audits.

The interview study was conducted with 19 auditing professionals in the widest sense to explore their understanding of algorithmic audits and how they aim to create users' trust in their audits and

in AI [7]. Currently, the AI auditing space is characterized by information asymmetry which can lead to a decreased value of algorithmic audits for users. The interviewed stakeholders indicated that they lack information on how much users' and the general public know about their audits and what their perspectives on the audits are. In the interviews, they suggested key elements for rebuilding trust among users and recommend measures like monitoring of auditors or improved communication about AI audits. A common concern among the interviewees which is echoed by the literature is that there is a lack of common standards and regulation regarding algorithmic audits which leads to a situation compared to the "Wild, wild West" [8].

To understand what elements of algorithmic audits are important to users and how audits could be built to increase calibrated trust among users, the interview study was followed by a survey in the Human Resource sector. Users in this study were jobseekers or recent jobseekers on one side who are subjected to AI in the hiring process, and human resource managers and recruiters on the other side, who use AI tools for the application and hiring process. The research found that currently users have only very limited knowledge about algorithmic audits, but once they learn more about them, they find them important and would like to see them conducted often. The survey results emphasized that users' currently are not well informed about algorithmic audits, confirming the information asymmetry regarding auditing, but that there is a strong interest in learning more about them. It is also showing that users' would like to see audits performed on a frequent basis. This underlines the need for more research including the perspectives of all stakeholders in the design of algorithmic audits.

5 FUTURE WORK

Future work will concentrate on a more detailed data analysis of the conducted survey. In a future paper build on the survey, I will analyze, in a mixed-method approach using both qualitative and quantitative methods, how much the end-users of employment tools (jobseekers and HR-managers/recruiters) trust such audits and what elements are essential for their trust. The analysis will focus on what elements of audits users find important and how they would like to see audits communicated to the public. It will further examine whether there are differences between HR-professionals and jobseekers. In addition, it will explore whether users' general trust in AI and trust in HR-tools influences their opinions about algorithmic audits.

Further work for my dissertation will go beyond the HR sector, e.g. including algorithmic audits in finance or health, and will also include how regulation, standards like the NIST framework [9], and algorithmic audits can come together to build calibrated trust in users of AI tools. I aim to conduct additional empirical studies to learn more about users' experiences in other sectors, and how important and useful they find regulation and/or soft law approaches.

6 CONTRIBUTIONS

Research on algorithmic auditing is still in a nascent stage. While there is more and more interest in algorithmic audits in general, there is a lack of empirical studies regarding users of artificial intelligence. I am aiming to fill this gap. I plan to add to previous work by adding qualitative and quantitative research examining how users trust in artificial intelligence could be influenced by algorithmic audits, and how different hard law and soft law approaches can come together to be more effective. Through my research, I seek to include users' interests in the discussions regarding algorithmic audits, as they are the stakeholders directly affected by AI. Their perspectives could help shape a better design of algorithmic audits and other soft law approaches, and thus could help increase the trust in and acceptance of trustworthy AI by all. Research results could inform public policies regarding audits, standards and regulation, while also providing practical guidance for professionals in auditing. Intended venues for my research results are interdisciplinary journals and conferences, in addition to venues aimed at practitioners.

7 DOCTORAL CONSORTIUM

The CSCW doctoral consortium could be highly valuable for my second half of my PhD studies. My initial research focus on algorithmic hiring and algorithmic audits in the human resource sector provided a solid foundation for my dissertation. In the next step, I would like to expand the research to include other sectors beyond human resource that could benefit from algorithmic audits. In addition, I would like to research the interplay between audits and other soft law approaches, such as the NIST standard [9], and how regulation could be applied to make auditing more effective. Any insights in what areas might be most interesting to include in my dissertation, complementing my current and previous research would be appreciated.

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